Avnish

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Summary

Motivated Mechanical Engineering student with strong technical and analytical skills, seeking internship/full-time opportunities. Skilled in AutoCAD, Creo, ANSYS, and Fusion 360, with hands-on project experience and a published patent in renewable energy solutions. Passionate about sustainable innovation, product design, and applying engineering principles in real-world problem-solving.

Experience

Internship – Airports Authority of India (Jul-Aug 2025)

- Gained exposure to airport infrastructure: HVAC, power plants, runway friction testing, and fire safety systems.
- Learned procurement processes aligned with Make in India policy.
- Applied mechanical and electrical engineering concepts in safety-critical environments.

Education

B.Tech – Mechanical Engineering

Dronacharya Group of Institutions, Greater Noida | Aug 2022 – Aug 2026

Skills

- **Design & Analysis:** AutoCAD, Creo, Fusion 360, ANSYS Workbench
- **Programming & Tools:** Python, Vehicle Diagnostic Software
- **Core Engineering Areas:** Thermal Analysis, Structural Analysis
- **Languages:** English, Hindi

Certifications

- Engine Management Systems & Automotive Electronics
- Student Coordinator at the 2nd International Conference on Optimization Techniques in Engineering.
- INDIA DESIGN WEEK 2025 ,National-level 24-Hour Design Competition(ICT Academy)
- IC Engines & Gas Turbines
- Introduction to Modeling & Design for Manufacturing

Key Achievements

- Patent Holder – Small-Scale Vertical Axis Wind Turbine for urban clean energy generation.
- Student Coordinator – International Conference on Optimization Techniques in Engineering.

Projects

- Small-Scale Vertical Axis Wind Turbine – Designed & patented compact wind turbine for urban energy generation, emphasizing sustainability.
- Steady-State Thermal Analysis (Cylindrical Model) – Conducted thermal analysis using ANSYS to study conductivity and gradients.
- Static Structural Analysis of Truss Structure – Modeled and analyzed truss structures for force reactions & stress distribution.
- Thermal Gradient Analysis (3D Model) – Performed 3D thermal analysis using ANSYS to visualize heat concentration zones.